Fiscal Year:	FY 2011	**Task Last Updated:**	FY 11/02/2010
PI Name:	Nelson, Gregory A. Ph.D.		
Project Title:	Epigenetic Control of Radiogenic Damage Processing in C. elegans		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92350-1700	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	01/01/2008	**End Date:**	06/29/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/29/2012 per A. Chu/ARC and NSSC (Ed., 10/31/11) NOTE: Received NCE until 12/31/2011, per NSSC info (10/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Braby, Leslie (Texas A&M University) Ford, John (Texas A&M University)		
Grant/Contract No.:	NNX08AC79G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Levels of expression of specific genes in individuals are determined by the particular combination of alleles that they carry and the regulatory environment of cells expressing those genes. Molecular pathways and mechanisms that detect and respond to damage from ionizing radiation are thus subject to modulation by external processes acting upon the genes expressed in the irradiated cells. These mechanisms constrain and determine the final levels of gene activity. These epigenetic mechanisms may be cell autonomous or operate at the tissue and organism level by processing information from cytokines, growth factors, hormones, and extracellular matrix. The range of gene expression patterns driven by the combinatorial actions of these epigenetic regulatory mechanisms ultimately determines the inter-individual variability of responses to radiation exposure. We propose to address these issues using a simple animal model, the nematode C. elegans, in which genotoxicity (chromosome aberrations) can be quantified in vivo in individual intestine cells whose tissue location and cell lineage are invariant. We have shown that individual cells in this 20-cell tissue respond to genotoxic damage in their neighbors and that their radiosensitivities appear to be modulated by signals originating in an adjacent tissue, the immature gonad. To identify the mechanisms underlying these cell-cell and tissue-tissue interactions we will employ a targeted screening protocol based on RNA interference methods to determine which of approximately 250 candidate genes alter the tissue architecture, cell-cell communication properties, DNA repair capacity, and important regulatory signal cascades which we believe are likely to modulate the expression and transmission of the genotoxic damage. This will be complemented by targeted irradiations of individual cells and cell groups with microbeam and collimated alpha particles to determine the range and topology of the damage transmission between cells.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	Low doses of ionizing radiation in the form of natural radioactivity, cosmic rays and medical diagnostic procedures comprise the background exposure levels for the general population. Occupational exposures may add to this dose and the sum of the two will establish health risks from radiation exposure. Risks from radiation exposure have been extrapolated from data on atomic bomb survivors, individuals exposed during radiation accidents, and older unregulated medical exposures. Regulatory limits have employed the linear no-threshold model for extrapolating risks to low doses but recent experimental findings suggest significant deviations from linearity at low doses such that the uncertainties in risk estimates must be re-evaluated. One of the important types of biological responses that may explain the deviations are those termed non-targeted effects. These are responses in which cells not directly exposed to radiation may sustain injury due to perturbations in their environment or transfer of damaging chemical species created in the irradiated cells. This task is focussed directly on identifying the genes and pathways that regulate non-targeted damage formation. The high genetic homology between C. elegans and mammalian gene networks should enable predictions of homologous signaling pathways that are operational in mammals, including humans. This should, in turn, improve the understanding of the role of non-targeted effects in the low radiation dose environment and suggest mitigation strategies.		

Task Progress:	As reported in last year's annual report, technical progress with respect to individual cell irradiations was limited by technical difficulties at the Texas A&M University microbeam facility. Meanwhile the genetic aspects of the project have proceeded at a reduced rate. In the spring of 2010 the facility was restored to normal operational status and experiments were immediately scheduled. Successful completion of the first round of experiments in July gave our team sufficient confidence that the project could be completed in collaboration with Texas A&M. Therefore a no-cost extension was requested to enable a multiple session experimental campaign and NASA granted this request. Genetic Analysis: Odds ratio statistical analysis reflecting conditional probabilities of all cell-cell combinations was applied to all radiation exposures including historical exposures and indicated a drop off of the "bystander" signaling intensity with distance but showing a weak influence by radiation quality. Analysis of results from genetic experiments in the dose range giving 20 to 50% bridge frequencies identified several classes of interest. So far, genes in the non-homologous end joining (NHEJ) DNA repair pathway have been implicated in repairing damage leading to anaphase bridge whereas 9-1-1 response and homologous repair genes are not. Surprisingly, the atm-1 and atl-1 genes that serve as master regulators for DNA damage detection did not seem to be required. The biggest surprise was the observation that inhibiting trt 1, a telomerase reverse transcriptase, reduces bridge frequency. We expected the opposite: that improper telomere maintenance would lead to chromosome fusions in conjunction with the bridge-break-bridge cycle and result in higher frequencies bridges. Inhibition of gap junction genes expressed in the gut so far seem to limit the bridge frequency patterns but do not abrogate the signaling. Genes associated with the insulin/insulin-like signaling pathway appear to enhance overall bridge frequency and are linked to the transcription factor daf-16. In particular mutations in the mev 1and smk-1 genes strongly potentiate bridge frequency. We are currently testing several integrin mediated signaling components and genes regulating microRNA metabolism because as many as 19% of all worm genes may be under miRNA regulation. Targeted irradiations: In the past year a new ImperX 2 Mp camera has been installed and tested on the microbeam system. Filters and sliders were obtained for the Axiomat microscope core and FITC filters purchased. Laboratory modifications required to obtain a biohazard level 2 permit were completed so that the recombinant worm strains needed for GFP targeting of nuclei could be utilized. Two irradiation runs were carried out with wild-type worms and over 300 worms were irradiated. Preliminary calculations, born out by the results of the experimental runs in July and October, indicate that a particle fluence of about 90 per cell nucleus (approximately 20 Gy) would provide a strong DNA damage signal. To calibrate the particle fluence rate in terms of the Faraday cup reading a solid state detector is placed over the collimator exit and the count rate is determined. In a typical calibration the particle fluence rate is 400 per second when the Faraday cup reading is 5 x 10-10 amps. In this case the exposure time to deliver 90 protons to a cell nucleus is 0.225 s. If the accelerator beam current varies the exposure time can be adjusted appropriately. Using this approach it has been possible to reduce the time required to irradiate an individual to as little as 3 minutes. To date we have irradiated 144 (July) +161 (October) = 305 animals with targeted local doses of either 20 or 10 Gy. Targets have included cells in intestinal rings 1 through 6, inclusive, and the cytoplasm between cells 3 & 4. The current targeting method requires real time distance measurements referenced from the anatomical site in the pharynx ("grinder") to the transverse plane containing the desired cell nuclei. The smooth contour of the body midline introduces length measurement errors such that targeting beyond intestinal ring 6 would be inaccurate. Thus use of the transgenic strain expressing GFP fluorescence in intestinal cells is now required and is expected to be implemented in a scheduled January experimental campaign. The results from these experiments are allowing us to understand the direction and range of signaling as well as the logic for each cell in terms of its ability to send and/or transmit signals to and from its neighbors. Combined with alpha particle irradiations we have also tentatively concluded that in the continuous epithelial sheet of the GI tract there is a compartment boundary at the pharyngeal-intestinal valve through which signals do not pass. The cellular/biochemical basis of such a boundary is a matter of speculation at this point.
Bibliography Type:	Description: (Last Updated: 03/26/2026)
Abstracts for Journals and Proceedings	Jones T, Ortloff L, Perez C, Nelson G. "In vivo Bystander Signaling in the Nematode C. elegans." 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, New York, May 16 – 19, 2010. Program and abstracts. Poster C-14. 21st Annual NASA Space Radiation Investigators' Workshop, May 16-19, 2010. , May-2010
Abstracts for Journals and Proceedings	Jones T, Ortloff L, Perez C, Braby L, Ford J, Nelson G. "The Search for Genes Regulating Bystander Signaling in the Nematode C. elegans." 56th Annual Meeting of the Radiation Research Society, Maui, Hawaii, Sept 25 – 29, 2010. Abstract Book. 56th Annual Meeting of the Radiation Research Society, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=9e011f35-da29-44e9-81ca-4b52cf617fb4&cKey=49b16fb4-85b8-458b-b706-06cca11af902&mKey=5fb93787-2553-4b3b-a915-8c1ac71f127a , Sep-2010
Articles in Other Journals or Periodicals	Nelson G, Braby L, Ford J. "Microbeam Investigations with the Nematode C. elegans." Texas A&M Dept Nuclear Engineering online newsletter, October 2010. , Oct-2010